

Carl Matthew Z. Reyes

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Professional Summary

Computer Technology student with practical experience in embedded systems development, IoT integration, hardware troubleshooting, and web application development. Experienced in Arduino programming, ESP32 microcontrollers, sensor integration, Python, Flask, SQL, and database management, with a focus on developing SoilSense, a TinyML-based soil analysis and crop recommendation system. Proficient in Figma, Canva, VS Code, and Microsoft Excel, with additional experience in multimedia production, event organization, and collaborative technology projects.

Education

Bulacan State University, **Malolos, Bulacan**
Bachelor of Industrial Technology Major in Computer Technology (Graduating) 2023 - 2027

Guiguinto National Vocational High School **Santa Maria, Bulacan**
Information and Communications Technology (ICT) - (With Honors) 2021 - 2023

Leadership & Extracurricular Experience

Bulacan Tech Expo (BTX), Vista Mall Malolos - Organizer Oct 2024– May 2025
Participated in event organization and coordination for a technology-focused expo, supporting event activities and participant engagement.

Technical Toolkit & Skills

Embedded Systems & IoT
Arduino programming, ESP32/ESP32-S3 microcontrollers, sensor integration, serial communication (UART/RS485), hardware prototyping, circuit troubleshooting
Web Development & Databases
Python, Flask, HTML, CSS, JavaScript fundamentals, SQL, SQLite, REST API integration, database management
UI/UX & Digital Tools
Canva, Visual Studio Code, Microsoft Excel, Word, interface design, wireframing, prototyping, digital content creation
Technical Projects & Core Skills
IoT system development, TinyML fundamentals, sensor data analysis, system integration, problem-solving, critical thinking, teamwork, adaptability, communication, time management

Academic & Portfolio Projects

CIT Computer Laboratory Inventory System — UI/UX Designer / Project Team
Designed a Figma-based admin dashboard prototype for monitoring computer laboratories, device inventory, room status, maintenance requests, and role-based access for Student Assistant, Professor, and Technician users. Applied interface hierarchy, consistent components, navigation, spacing, and interactive prototype flows.
AgriSense — UI/UX Designer / QA Tester & IoT Developer
Contributed to the development of an integrated soil-monitoring and crop- recommendation system combining ESP32-S3 hardware, soil sensors, and a Flask- based web application. Supported UI/UX design, hardware- software integration, and quality assurance for real-time soil data acquisition, sensor communication, Wi-Fi data transmission, soil fertility scoring, and KNN- based crop recommendations. Worked with a RS485 Modbus NPK soil sensor, pH sensor, temperature and humidity monitoring, PostgreSQL database management, and an interactive dashboard for soil records and agricultural learning.

References

Mrs. Chelsar R. Fajardo
Manager- Bounty Fresh Food, Inc
09653606060